

SAHIL HEGDE

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EDUCATION

University of California, Los Angeles Los Angeles, CA	2021 - present
<i>Ph.D. Astronomy & Astrophysics, Advisor: Steve Furlanetto</i>	
<i>M.S. Astronomy & Astrophysics</i>	2021 - 2023
Columbia University New York, NY	2017 - 2021
<i>B.A. Physics & Mathematics, Cum Laude, Advisors: Greg Bryan, Zoltan Haiman</i>	
<i>Other Experience:</i>	
CIRTL Teaching-as-Research (TAR) Certification	2026
ISEE Professional Development Program (PDP) participant and team leader	2025-2026

RESEARCH APPOINTMENTS

NSF & NASA FINESST Graduate Research Fellow / Graduate Student Researcher	2021-present
<i>University of California, Los Angeles</i>	
Undergraduate Research Assistant	2018-2021
<i>Columbia Univ. & Center for Computational Astrophysics; advisors: Greg Bryan, Zoltan Haiman, Shy Genel</i>	
Computational Physics (XCP) Summer Intern	Summer 2019
<i>Los Alamos National Laboratory; advisors: Brandon Wiggins, Chris Fryer</i>	
Undergraduate Research Assistant	Summer 2018
<i>Stanford University; advisors: Monica Bobra, Phil Scherrer</i>	

AWARDS AND HONORS

· UCLA Dissertation Year Fellowship (\$30,000)	2026-2027
· UCLA Michael A. Jura Fellowship (<i>for outstanding research and department citizenship</i> ; \$10,000)	2026
· UCLA Center for Developing Leadership in Science Fellowship / Leadership Facilitator (\$4000)	2025-2026
· Future Investigators in NASA Earth & Space Sciences Technology Fellowship (\$150,000)	2023-2026
· National Science Foundation Graduate Research Fellowship (\$159,000)	2023-2026
· UCLA Center for Developing Leadership in Science Early Career Fellowship (\$1000)	2024-2025
· UCLA Grad Slam Semifinalist (\$500)	2024
· UCLA Graduate Dean's Scholar Award (\$34,000)	2021
· Columbia College Dean's List	2018-2021
· Los Alamos National Lab Computational Physics Summer Fellowship	2019
· National Merit Scholarship	2017
· Second Place in Physics and Astronomy at Intel International Science and Engineering Fair	2017
· Siemens Competition Semifinalist	2017
· Saratoga Rotary Scholarship	2017
· Elks Lodge Scholarship	2017

SELECTED PUBLICATIONS

6 first-authored publications, 100 total citations (according to ADS as of October 4, 2026)

9. S. Hegde, G. Sun, J. B. Muñoz, S. R. Furlanetto; 2026; *Searching for Population III stars with line intensity mapping cross-correlations*; arXiv (under review in PRD)
8. S. Hegde, S. R. Furlanetto, S. Naoz; 2026; *Do we understand the star formation history of the universe?*; arXiv (under review in ApJ)
7. S. Hegde, S. R. Furlanetto; 2025; *Efficient semi-analytic modelling of Pop III star formation from Cosmic Dawn to Reionization*; Open Journal of Astrophysics
6. S. Hegde, M. M. Wyatt, S. R. Furlanetto; 2024; *A hidden population of active galactic nuclei can explain the overabundance of luminous $z > 10$ objects observed by JWST*; JCAP
5. S. Hegde, S. R. Furlanetto; 2023; *A self-consistent semi-analytic model for Population III star formation in minihaloes*; MNRAS
4. B. L. Lewis, Astrobites Collaboration, et al. (incl. S. Hegde); 2025; *Improving undergraduate astronomy students' skills with research literature via accessible summaries: An exploratory case study with Astrobites-based reading assignments*; Physical Review Physics Education Research
3. S. Knabel, et al. (incl. S. Hegde); 2023; *Modelling strong lenses from wide-field ground-based observations in KiDS and GAMA*; MNRAS
2. S. Hegde, G. L. Bryan, S. Genel; 2022; *Reorientation Rates of Structural and Kinematic Axes in Simulated Massive Galaxies and the Origins of Prolate Rotation*; ApJ
1. S. Hegde, M. G. Bobra, P. Scherrer; 2018; *Classifying Signatures of Sudden Ionospheric Disturbances*; RNAAS

ACADEMIC PRESENTATIONS

25. *Evaluating the consistency of fundamental galaxy scaling relations in the JWST era*; Galaxy formation and evolution in Southern California (GALFRESCA) 2026; September 2026; Irvine, CA; contributed talk
24. *Constraining the demographics of the first stars with line intensity mapping*; Line Intensity Mapping 2026; August 2026; Pasadena, CA; contributed talk
23. *Establishing a census of star formation across cosmic time with Roman*; Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman; July 2026; Pasadena, CA; contributed talk
22. *Understanding star formation in the first billion years with flexible galaxy models*; Kamionkowski group meeting; May 2026; Baltimore, MD; invited blackboard seminar
21. *Tracking sites of low-metallicity star formation with flexible galaxy models*; STScI Spring Symposium, The Metal-Poor Frontier: Understanding Low-Metallicity Stars and Galaxies Across Cosmic Time; May 2026; Baltimore, MD; contributed talk
20. *Understanding star formation in the first billion years with flexible galaxy models*; UMD Special Seminar; May 2026; College Park, MD; invited seminar
19. *Testing assumptions in high-redshift galaxy formation for models of the global 21-cm signal*; Global 21-cm workshop VIII; September 2025; Pasadena, CA; contributed talk
18. *Simple models for the first stars and galaxies*; Sierra Conference 2025; Yosemite, CA; poster presentation
17. *Identifying imprints of the first stars across cosmic time*; First galaxies: the building blocks of galaxies across cosmic time; April 2025; Oxford, UK; contributed talk
16. *Self-consistent semi-analytic modelling of the first stars and their observational signatures*; Galaxy formation and evolution in Southern California (GALFRESCA) 2024; September 2024; Pasadena, CA; contributed talk

15. *A semi-analytic model for Pop III star formation*; KITP Cosmic Dawn Revealed by JWST: The physics of the first stars, galaxies, and black holes; August 2024; Santa Barbara, CA; contributed talk
14. *AGNs and the high- z UVLF*; Sierra Conference 2024; August 2024; Yosemite, CA; poster presentation
13. *The roles of AGNs at high- z and the reionization era*; NORDITA Cosmic Dawn at High Latitudes program; June 2024; Stockholm, Sweden; invited discussion
12. *A semi-analytic model for Pop III star formation*; NORDITA Cosmic Dawn at High Latitudes conference; June 2024; Stockholm, Sweden; contributed talk
11. *Black holes and the UV luminosity function*; Center for Computational Astrophysics galaxies group meeting; May 2024; New York, NY; invited seminar
10. *A semi-analytic model for Pop III star formation*; First stars VII; May 2024; New York, NY; contributed talk
9. *A semi-analytic model for Pop III star formation*; Global 21-cm workshop VI; September 2023; Trieste, Italy; contributed talk (remote)
8. *The minimum halo mass for Pop III star formation*; NESS group meeting; April 2023; Boulder, CO; invited seminar (remote)
7. *A semi-analytic model for high-redshift star formation*; Supersonic project workshop; March 2023; Los Angeles, CA; invited talk
6. *Radio astrophysics from the moon: signatures of the first stars*; American Astronomical Society 241st meeting; January 2023; Seattle, WA; poster presentation (remote)
5. *Predicting radio signals from the first stars*; NASA Exploration Science Forum; July 2022; Boulder, CO; poster presentation
4. *Pop III star formation in the presence of a Lyman-Werner background and stream velocity*; NESS group meeting; May 2022; Boulder, CO; invited seminar (remote)
3. *The origins of prolate rotation in simulated massive early-type galaxies*; Center for Computational Astrophysics galaxies group meeting; May 2021; New York, NY; invited seminar
2. *The time evolution of the misalignment between angular momentum and principal axis in simulated massive early-type galaxies*; American Astronomical Society 237th meeting; January 2021; poster presentation (remote)
1. *On the supermassive black hole-galaxy coevolution*; American Astronomical Society 229th meeting; January 2017; Grapevine, TX; poster presentation

TEACHING

Instructor & Course Designer, Astronomy Introductory Coding Workshop Winter 2024, 2025, 2026
Led and created an annual 10-week course for UCLA undergraduates to learn and practice fundamental python research skills. Los Angeles, CA

Lead Instructor & Designer, Astronomy Live! Summer Observing Workshop Summer 2022-2026
Annual 9-week summer course for Los Angeles area high school students to learn the fundamentals of observational astronomy and python. Los Angeles, CA

Team Leader & Activity Designer, ISEE Professional Development Program 2025, 2026
Led design and creation of inquiry-based astronomy lab activities for community college and undergraduate students through the Institute for Scientist and Engineer Educators (ISEE). Los Angeles, CA

Graduate Teaching Assistant, UCLA Department of Physics and Astronomy Sept. 2021 - June 2022
ASTR 3: The Nature of the Universe (lower division astronomy lab) Los Angeles, CA
ASTR 115: Statistical Mechanics in Astronomy

ASTR 117: Radiative Processes & Fluid Dynamics in Astronomy

Math Help Room Facilitator, Columbia University
Assisted students for all upper division undergraduate math courses.

Jan. 2019 - May 2020
New York, NY

Undergraduate Teaching Assistant, Columbia University
MATH 2030: Ordinary Differential Equations

Jan. 2019 - May 2020
New York, NY

Math & Physics tutor, bitWise Academy
Tutored high school-level mathematics and physics courses.

2022-2024
Los Angeles, CA

MENTORSHIP

Research mentees:

- Adam Janik, UCLA undergraduate Spring 2025-present
Adam is quantifying the effects of stochastic merger histories in simple models of galaxy formation.
- Brijesh Molala, UCLA undergraduate Winter 2023-Spring 2024, Winter 2026-present
Brijesh is exploring the implications of physically-motivated bursty star formation models on galaxy luminosity functions and their clustering.
- Cassidy Klahn, UCLA undergraduate Spring-Summer 2025
Cassidy explored the effects of a fuzzy dark matter cosmology on the formation and abundance of high- z galaxies.
- Jacob Laurel, UCLA undergraduate (now UCI graduate student) Spring 2024-Summer 2025
Jake explored the effects of stochasticity in IMF sampling on feedback and bursty star formation histories in simple analytical and semi-analytical models of galaxy formation.
- Malachy Bloom, Carleton undergraduate 2023
Malachy explored the effects of a warm dark matter cosmology on the formation and abundance of high- z minihalos and galaxies.
- Roy Zhao, UCLA undergraduate (now Stanford graduate student) 2022
Roy worked on developing a framework to constrain the pop III IMF using transients associated with the first stars.

Graduate peer mentees:

- Marwa Selim (University of Florida)
- Sean Granados (UCLA)

Undergraduate mentees:

- Laura Hernandez (UCLA)
- Abigail Lynch (UCLA)
- Vikram Seenivasan (UCLA; now Cambridge grad.)
- Kevin Hong (UCLA; now UChicago grad.)
- Yan Sun (UCLA)
- Michael Karkar (UCLA)
- Joseph Lau (UCLA)

OUTREACH, ADVOCACY, AND PROFESSIONAL SERVICE

Reviewer
Nature Astronomy, Astrophysical Journal, Monthly Notices of the Royal Astronomical Society, Journal of Cosmology and Astroparticle Physics

2021-present

Astrobites <i>Author, Committee Chair (DEI, Recruitment), Admin committee</i>	Dec 2021-present
Lead graduate student representative; Physics graduate council representative <i>UCLA Department of Physics & Astronomy</i>	June 2024-present
Senior coordinator, planetarium presenter <i>UCLA Planetarium</i>	Fall 2023-present
Workshop lead <i>UCLA Astronomy Live! High School Summer Workshop</i>	May 2022-present
Co-founder & co-lead instructor <i>UCLA Astronomy Introductory Coding Workshop</i>	Jan 2024-present
Co-founder <i>ComSciCon Southwest</i>	Sept. 2024-Dec. 2025
Tutor <i>Prison Mathematics Project</i>	2025-present
Graduate Student Representative, Treasurer <i>UCLA Math & Physical Sciences Council</i>	Sept. 2022-present
Undergraduate peer mentor <i>UCLA Marginalized Identities in Physics and Astronomy</i>	Sept. 2022-present

PUBLIC TALKS AND WRITING

UCLA Grad Slam <i>Finding the first stars</i>	March 2024
Santa Barbara Astronomical Unit <i>The formation of the first stars in the universe</i>	July 2023
iTelescope <i>The first stars in the universe</i>	February 2023
UCLA Planetarium	Fall 2021-present

- Delivered regular assorted shows for general public, school groups, and other visitors
- Spearheaded collaborations with LA area comedians
- Organized shows celebrating particular identity groups and heritage months
- Developed climate justice show series

Astrobites

- N. Sanders, **S. Hegde**, *Nathan Sanders is Rewiring Democracy*, Oct. 13, 2025
- **S. Hegde**, *#BlackInAstro Experiences: Logan White*, June 18, 2025
- **S. Hegde**, *AAS Diversity Committee Wrapped 2024*, Jan. 5, 2025
- **S. Hegde**, S. Wong, *Book Review: Diversity, Equity, and Inclusion in Astronomy: A Modern History by Dr. Jörg Matthias Determann*, Aug. 11, 2024
- **S. Hegde**, *Interview series: Dr. Jorge Moreno*, Sept. 14, 2024
- **S. Hegde**, *Equitable access in an increasingly computational world*, Sept. 5, 2024
- **S. Hegde**, *A journey through space and time with #BlackInAstro*, June 15, 2024
- **S. Hegde**, M. Popinchalk, *To be the eclipsed: astronomy outreach in the prison system*, April 5, 2024
- **S. Hegde**, *Modeling the structure of the circumgalactic bathtub*, Feb. 3, 2024

- S. Hegde, *Peering at stellar streams through a myrmecological magnifying glass*, Dec. 28, 2023
- S. Hegde, *The Rainbow Village at AAS: Dr. Arianna Long and VanguardSTEM*, Dec. 11, 2023
- S. Hegde, *Carnivalgoing galaxies and a billion dollar guessing game*, Oct. 14, 2023
- S. Hegde, *A quantitative perspective on the impact of diversity in academia*, Sept. 28, 2023
- S. Hegde, *Interview series: Ronald López*, Aug. 25, 2023
- S. Hegde, *The terrible twos: the detection of a toddler black hole in the baby universe*, July 6, 2023
- S. Hegde, *Juggling black holes in star clusters*, May 24, 2023
- S. Hegde, *How to build an infrared luminosity function (at high redshifts)*, April 18, 2023
- S. Hegde, M. Popinchalk, *Book Review: Under Alien Skies by Dr. Phil Plait*, April 16, 2023
- S. Hegde, *#BlackInAstro Experiences: Dr. Chris Moore*, Feb. 24, 2023
- S. Hegde, *A grad student seat at the faculty hiring table*, Jan. 23, 2023
- S. Hegde, *Meet the AAS Keynote Speakers: Prof. Norman Murray*, Jan. 10, 2023
- S. Hegde, *Meet the AAS Keynote Speakers: Prof. Erin Kara*, Jan. 8, 2023
- S. Hegde, *Cracking the Code for Galaxy Spectra*, Nov. 26, 2022
- S. Hegde, *Moonspotting: An Indirect Channel to Observe Rocky Planet Moons*, Oct. 24, 2022
- S. Hegde, *Pulling Back the Curtain on the Formation of the First Supermassive Black Holes*, Aug. 25, 2022
- S. Hegde, *Adding Gal-ax-ies, It's Easy as One-Two-Three*, Aug. 1, 2022
- S. Hegde, *(almost) No Time (for stars) to Die*, June 27, 2022
- S. Hegde, O. Cooper, *Meet the AAS Keynote Speakers: Prof. Victoria Kaspi and Dr. Paul Scholz*, June 12, 2022
- S. Hegde, *A Missing Link Between High-Redshift Galaxies and Quasars*, May 11, 2022
- S. Hegde, *Women's History Month: Dr. Ekta Patel*, May 1, 2022
- S. Hegde, *Judging a Galaxy by its Cover*, Mar. 1, 2022
- S. Hegde, *#BlackInAstro Experiences: Dr. Gibor Basri*, Feb. 27, 2022
- S. Hegde, *RRow RRow RRow Your [Tele]scope, Gently Down the [Orphan] Stream*, Feb. 14, 2022